

UNIVERSITI TEKNOLOGI MARA

FACULTY OF COMPUTER AND MATHEMATICAL SCIENCES
COLLEGE OF COMPUTING, INFORMATICS AND MATHEMATICS

CSC649 - SPECIAL TOPICS IN COMPUTER SCIENCE

TITLE:

ASSIGNMENT 1:

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1.0 Introduction



Supercook is a popular cook apps recipe discovery platform designed to help users find meals based on the ingredients they currently have at home. By allowing users to input available items, the app generates matching recipes to maximize food utility and simplify meal planning. The app uses Natural Language Processing techniques by choosing specific ingredients to determine the meals based on what the user has. Therefore, several issues have to be solved where users may still struggle to decide how to turn leftover food into new meals. If this problem is not going to be solved, it can lead to food waste.

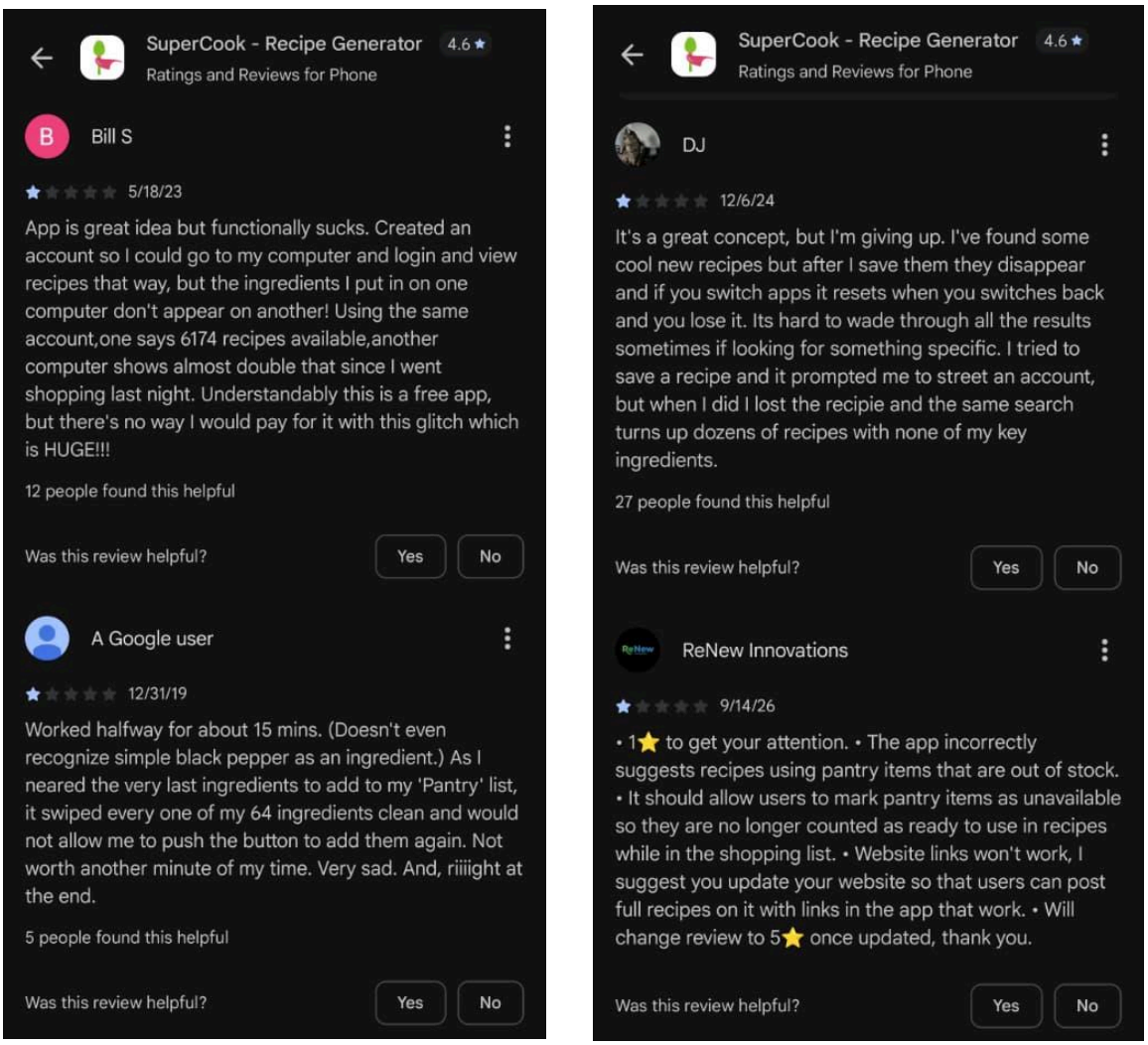
To effectively address these issues, it is important to understand both User Interface (UI) and User Experience (UX). UI refers to visual elements such as layout, typography, colours, and buttons. On the other hand, UX refers to how the interface feels, focusing on the user journey, ease of navigation, and overall efficiency.

This project applies Design Thinking and a human-centred used to solve complex problems through five core stages:

- **Empathize:** Research and understand user needs, pain points, and behaviors through observation and feedback.
- **Define:** Analyze research insights to clearly articulate core problem statements and user needs.
- **Ideate:** Brainstorm a wide range of creative solutions and potential feature improvements to address the defined problems.
- **Prototype:** Build experimental, low- or high-fidelity representations of proposed solutions to explore and refine ideas.
- **Evaluate:** Test the prototype with users to gather feedback, validate assumptions, and guide further iterations.

2.0 Empathize

Standard recipe systems rely heavily on rigid database matching and explicit exact-match filtering (Rodrigues et al., 2023). When a user enters arbitrary or sparse leftover ingredients, traditional algorithms struggle to generate flexible options, often requiring the user to supply numerous additional ingredients. This rigid constraint leads to "recipe fatigue" and causes users to abandon leftover utilization in favor of buying new food (Rodrigues et al., 2023).



1.0 User's review on Supercook app from Google App Store

This image highlighted users feedback from App Store reviews several users having problems with UI/UX repeatedly functional glitches in current mobile implementations. A major issue has been discussed where severe data synchronization across devices. For example, users report saving ingredients on a laptop only to find them completely missing when logging in with the same account

on a phone. Additionally, users experience frequent state resets where saved or planned recipes and menus disappear when switching between applications. Finally, there is widespread frustration with rigid recipe matching algorithms that incorrectly suggest recipes requiring out-of-stock ingredients, leading to wasted effort and user dissatisfaction.

3.0 Define

Based on the limitations that appear in current Supercook apps. There are three core problem statements:

1. Home cooks utilizing leftover ingredients need reliable data synchronization because frequent app resets and disappearing saved recipes cause data loss and disrupt meal planning.
2. Users managing leftover items need flexible ingredient matching and accurate pantry tracking because rigid filtering and suggestions using out-of-stock items lead to increased food waste.
3. First-time app users need an intuitive guideline flow because excessive friction during sign-up and overwhelming search results discourage users from effectively discovering recipes.

4.0 Ideate

Idea 1	
What is the improvement?	Implement automatic cloud-sync and local offline caching for saved recipes and pantry lists.
Which pain point does it solve?	To prevent data loss when the user saves the recipe.
Why is it better than the current design?	Ensures user data is preserved instantly across devices without relying solely on manual account logins.

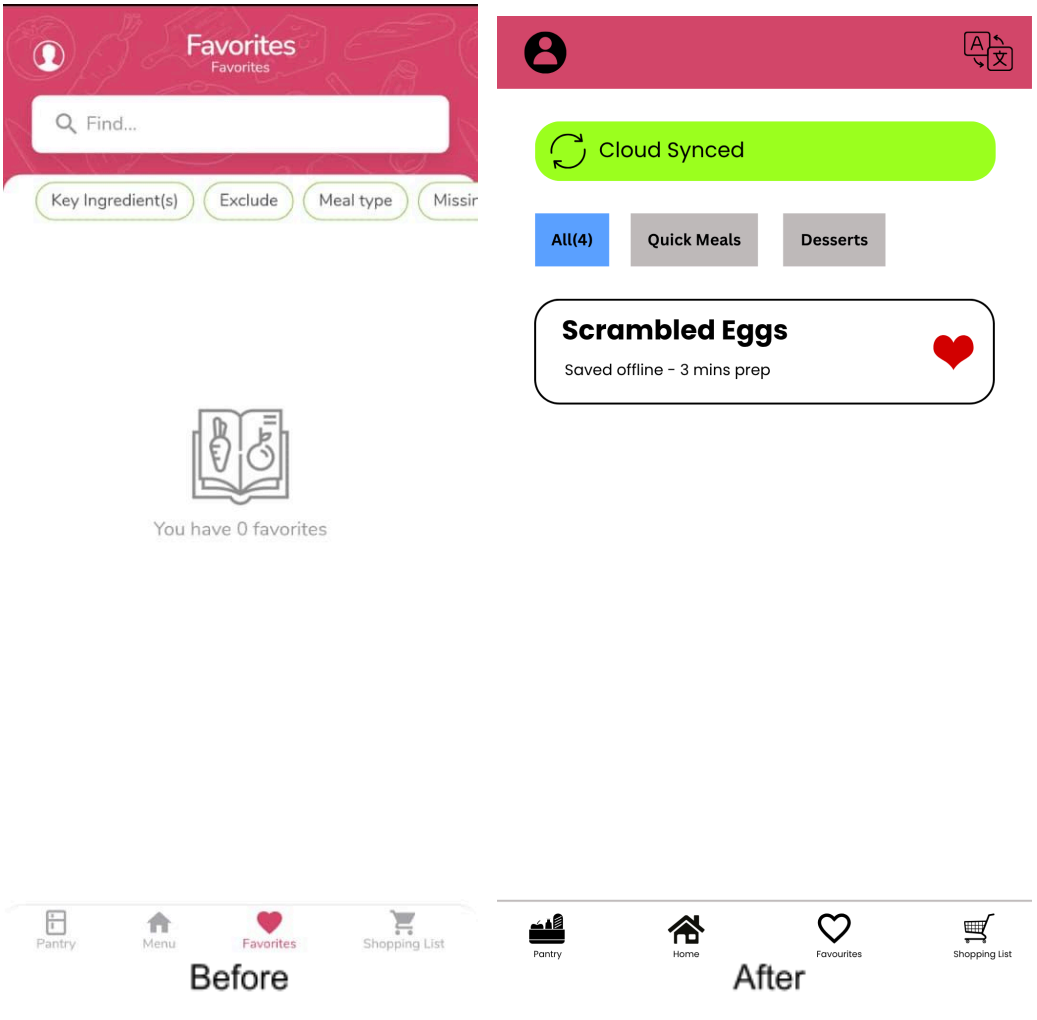
Idea 2	
What is the improvement?	Add an intelligent substitute recommender and a toggle to mark items as out-of-stock.
Which pain point does it solve?	Rigid ingredient matching & out-of-stock items
Why is it better than the current design?	Gives users flexibility when ingredients are missing, reducing food waste and recipe frustration.

Idea 3	
What is the improvement?	Introduce a streamlined "guest mode" onboarding and a simplified ingredient filter search bar.
Which pain point does it solve?	UI/UX friction & overwhelming results
Why is it better than the current design?	Reduces sign-up friction and helps users find relevant recipes in fewer steps.

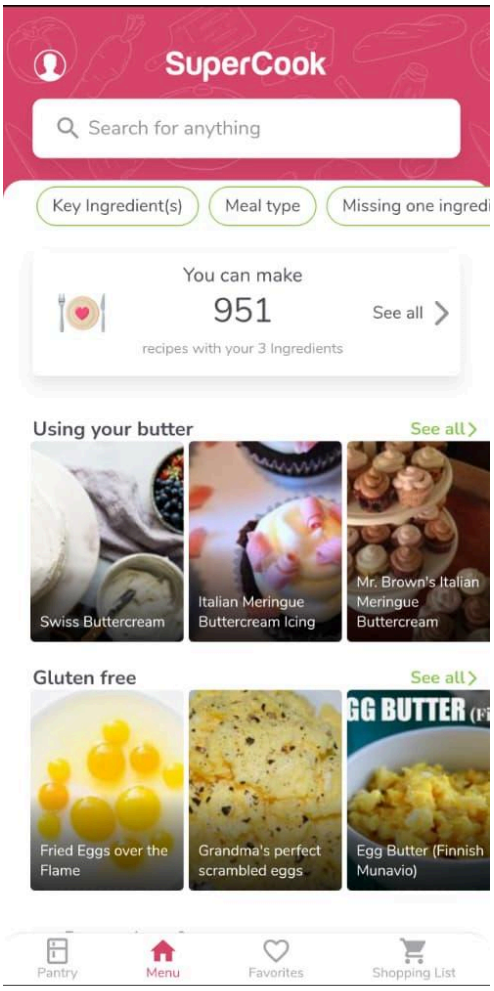
5.0 Prototype

The prototype addresses the core issues identified during the research and ideation phases. First, it introduces automated cloud synchronization with local offline caching, ensuring users never lose their saved recipes or pantry data when switching devices or offline. Second, an intelligent ingredient recommender along with clear stock toggles allows users to substitute missing ingredients and track inventory accurately, preventing food waste. Lastly, a streamlined guest mode onboarding and a unified search interface significantly reduce initial friction and make finding relevant recipes effortless.

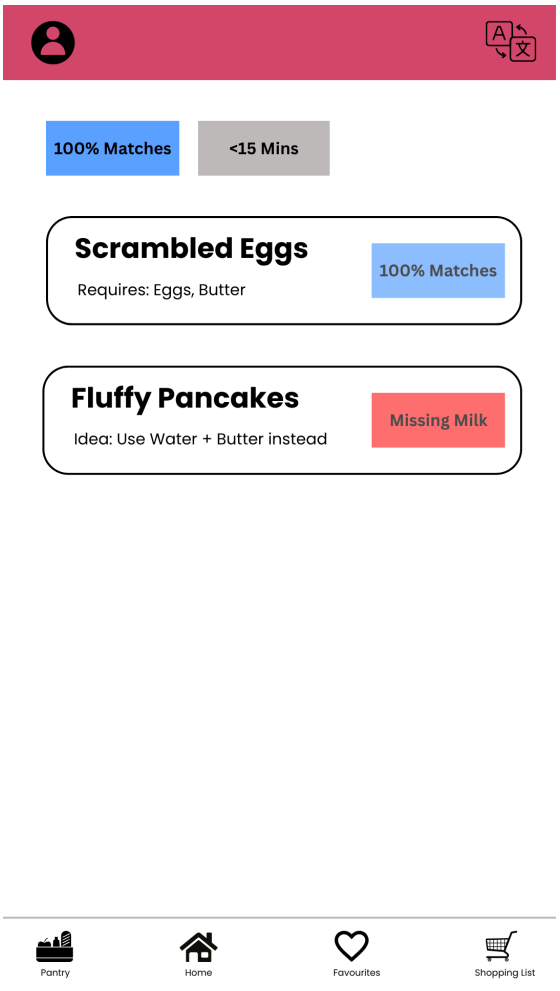
1. Cloud Sync & Offline State Indicator



2. Smart Ingredient Management & Out-of-Stock Toggle

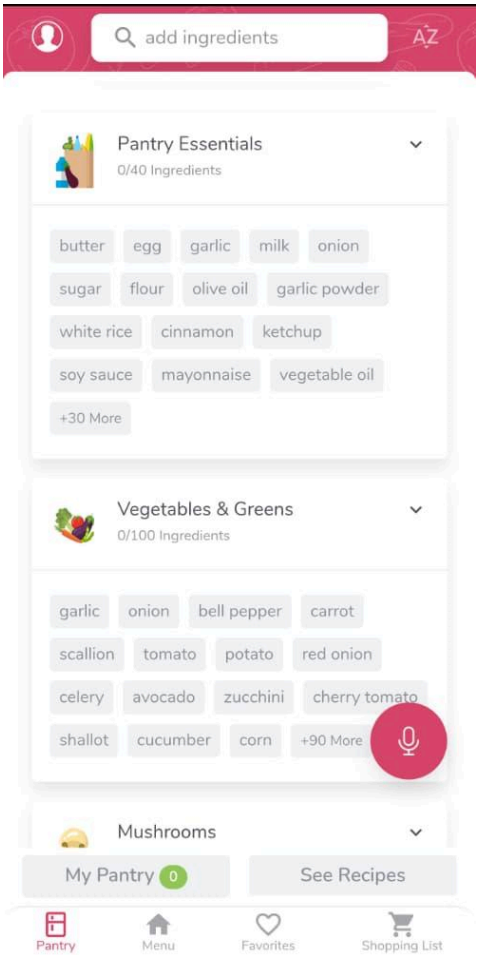


Before

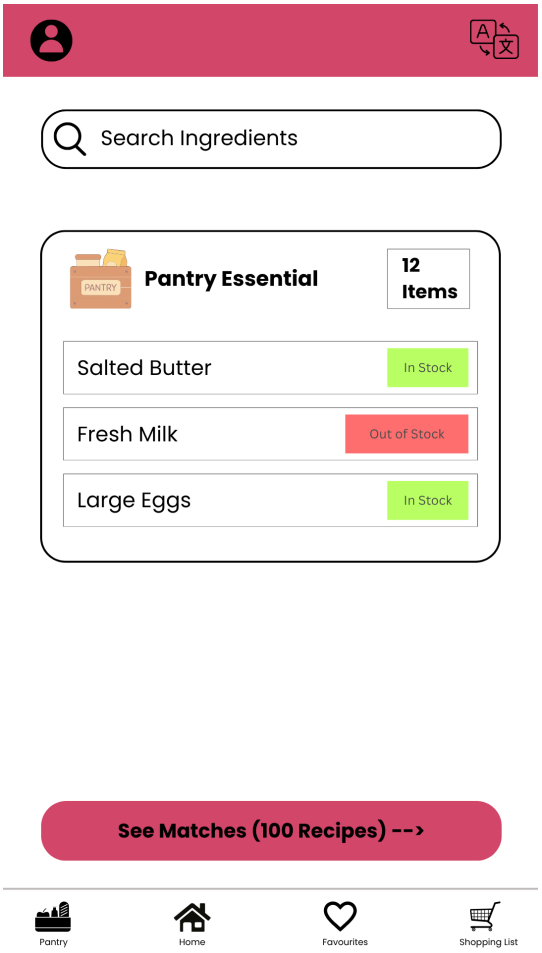


After

3. Simplified Search Filters



Before



After

6.0 Evaluate

The evaluation phase assessed the redesigned Supercook prototype through targeted usability testing and feedback sessions with five participants. Users performed core tasks, including setting up an initial pantry inventory, toggling item availability, searching for missing ingredient substitutes, and syncing saved recipes across multiple test devices.

What Worked Well

Usability testing confirmed that automatic cloud synchronization and offline caching successfully resolved data loss issues, ensuring saved recipes and pantry lists remained intact across device state changes. Additionally, users found the intelligent substitute recommender and out-of-stock toggle highly effective in reducing friction and facilitating leftover ingredient utilization. The simplified guest onboarding and streamlined search filters further minimized initial navigation barriers for new users.

What Can Be Improved

Future recommendations include integrating localized dietary preference filters and expanding smart ingredient substitution suggestions to support a wider range of regional cuisines.

Personal Reflection

Working on this project highlighted the importance of user-centered design in turning everyday cooking challenges into seamless digital experiences. Iterating on the Supercook interface demonstrated how addressing small UI frictions such as unexpected state resets or rigid matching rules can significantly improve usability and promote sustainable home habits like minimizing food waste.